

Interacting Multipoles of Rank 2 and 2:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}^A\Theta_{\gamma\delta}^B = R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{l} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta(\beta, \delta) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta(\alpha, \delta) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta(\alpha, \gamma) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \beta) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \end{array} \right] \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}^A\Theta_{\gamma\delta}^B = R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{aligned} &+105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\beta\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\beta\delta}^B \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\delta}^B \\ &-15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\gamma\delta}^B \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\beta\beta}^B \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\beta}^B \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\beta}^B \end{aligned} \right] \quad (2a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\gamma}^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\beta\gamma}^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\beta\delta}^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\gamma}^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\delta}^B \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\gamma\delta}^B \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\beta\beta}^B + 3 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\beta}^B \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\beta}^B \end{array} \right] \quad (2b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\gamma}^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\beta\gamma}^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\beta\gamma}^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\gamma}^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\gamma}^B \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\beta \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\beta\gamma}^B \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\beta\beta}^B + 3 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\beta}^B \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\beta}^B \end{array} \right] \quad (2c)$$

$$\xRightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\delta}^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\gamma\gamma}^B \\ -30 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\beta\gamma}^B \\ -30 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\gamma}^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\beta\gamma}^B \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\beta\beta}^B + 6 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\beta}^B \end{bmatrix} \quad (2d)$$

Simplify R:

$$T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^A \Theta_{\gamma\delta}^B = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^A \cdot \Theta_{zz}^B \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{zz}^A \cdot \Theta_{\gamma\gamma}^B \\ -30 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\beta}^A \cdot \Theta_{z\beta}^B \\ -30 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha}^A \cdot \Theta_{z\alpha}^B \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{zz}^B \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\beta\beta}^B + 6 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\beta}^B \end{bmatrix} \quad (3a)$$

$$\xRightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz}^A \cdot \Theta_{zz}^B - 15 \cdot R_z^4 \cdot \Theta_{zz}^A \cdot \Theta_{\gamma\gamma}^B \\ -30 \cdot R_z^4 \cdot \Theta_{z\beta}^A \cdot \Theta_{z\beta}^B - 30 \cdot R_z^4 \cdot \Theta_{z\alpha}^A \cdot \Theta_{z\alpha}^B \\ -15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{zz}^B + 3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\beta\beta}^B \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\beta}^B \end{bmatrix} \quad (3b)$$

$$\xRightarrow{\text{reduced indices}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz}^A \cdot \Theta_{zz}^B - 15 \cdot R_z^4 \cdot \Theta_{zz}^A \cdot \Theta_{\alpha\alpha}^B \\ -30 \cdot R_z^4 \cdot \Theta_{z\alpha}^A \cdot \Theta_{z\alpha}^B - 30 \cdot R_z^4 \cdot \Theta_{z\alpha}^A \cdot \Theta_{z\alpha}^B \\ -15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{zz}^B + 3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\beta\beta}^B \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\beta}^B \end{bmatrix} \quad (3c)$$

$$\xRightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz}^A \cdot \Theta_{zz}^B - 15 \cdot R_z^4 \cdot \Theta_{zz}^A \cdot \Theta_{\alpha\alpha}^B \\ -60 \cdot R_z^4 \cdot \Theta_{z\alpha}^A \cdot \Theta_{z\alpha}^B - 15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{zz}^B \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\beta\beta}^B + 6 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^A \cdot \Theta_{\alpha\beta}^B \end{bmatrix} \quad (3d)$$

Simplify Quadrupoles:

$$T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^A \Theta_{\gamma\delta}^B = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz}^A \cdot \Theta_{zz}^B - 15 \cdot R_z^4 \cdot \Theta_{zz}^A \cdot \Theta_{\alpha\alpha}^B \\ -60 \cdot R_z^4 \cdot \Theta_{z\alpha}^A \cdot \Theta_{z\alpha}^B - 15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{zz}^B \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\beta\beta}^B + 6 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\alpha\alpha}^B \end{bmatrix} \quad (4a)$$

$$\xRightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +45 \cdot R_z^4 \cdot \Theta_{zz}^A \cdot \Theta_{zz}^B \\ -15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\beta\beta}^B \\ -15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{zz}^B \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\beta\beta}^B \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\alpha\alpha}^B \end{bmatrix} \quad (4b)$$

Exploit Traceless Conditions:

$$T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^A \Theta_{\gamma\delta}^B = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +45 \cdot R_z^4 \cdot \Theta_{zz}^A \cdot \Theta_{zz}^B \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^A \cdot \Theta_{\alpha\alpha}^B \end{bmatrix} \quad (5)$$

Multiply Out Sum:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}^A\Theta_{\gamma\delta}^B = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +45 \cdot R_z^4 \cdot \Theta_{zz}^A \cdot \Theta_{zz}^B \\ +6 \cdot R_z^4 \cdot \Theta_{xx}^A \cdot \Theta_{xx}^B \\ +6 \cdot R_z^4 \cdot \Theta_{yy}^A \cdot \Theta_{yy}^B \\ +6 \cdot R_z^4 \cdot \Theta_{zz}^A \cdot \Theta_{zz}^B \end{bmatrix} \quad (6a)$$

$$\xRightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +51 \cdot R_z^4 \cdot \Theta_{zz}^A \cdot \Theta_{zz}^B \\ +6 \cdot R_z^4 \cdot \Theta_{xx}^A \cdot \Theta_{xx}^B \\ +6 \cdot R_z^4 \cdot \Theta_{yy}^A \cdot \Theta_{yy}^B \end{bmatrix} \quad (6b)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}^A\Theta_{\gamma\delta}^B = \begin{bmatrix} +\frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{zz}^A \cdot \Theta_{zz}^B \\ +\frac{2}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^A \cdot \Theta_{xx}^B \\ +\frac{2}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^A \cdot \Theta_{yy}^B \end{bmatrix} \quad (7)$$